Vinayak Gupta

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About

Software Engineer and ML Engineer with experience building Python-based Generative AI and Large Language Models (LLMs) applications for enterprise use cases. Delivered Retrieval-Augmented Generation (RAG) solutions, standardized LLMOps practices, and production-grade data and AI workflows across Azure, Docker, Kubernetes, SQL, and GitHub Actions environments. Brings strong problem solving, experimentation, and end-to-end delivery experience across data ingestion, preprocessing, model evaluation, and deployment.

Skills

Programming Languages	Python | JavaScript | Go | C#
Generative AI & ML	Large Language Models (LLMs) | Retrieval-Augmented Generation (RAG) | Multi-Agent Systems (Agent2Agent / A2A) | Autonomous AI Agents (Claude Code) | Azure OpenAI | Prompt Flow | TensorFlow | Random Forest | SVM | CNN | RNN
Data Engineering	SQL | NumPy | Databricks | Azure Data Factory | Power BI | Excel | Apache Spark
Cloud / LLMOps	Azure | Azure DevOps | Azure ML SDK | Microsoft Agent Framework | Azure Container Apps | Cosmos DB | OpenTelemetry | Docker | Kubernetes | Git | GitHub Actions | Terraform | Bicep
Web Technologies	Django | Flask | ReactJS | Markdown
Soft Skills	Collaboration | Communication | Problem Solving | Logical Thinking

Work Experience

SimCorp — Noida, Uttar Pradesh
Senior Software Engineer - MLOps — July 2024 - Present

- Architected and deployed a zero-configuration AI code reviewer across all Azure DevOps repositories. Triggered via webhooks, the system enriches PR diffs with work item acceptance criteria and file context to output structured, severity-ranked reviews. Designed with strict security governance (read-only access) and smart comment fingerprinting to update existing discussion threads idempotently without spamming PRs.
- Contributed to a production multi-agent orchestration system (Microsoft Agent Framework, .NET, Azure OpenAI), including multi-agent planning, an MCP gateway for internal APIs, and evaluation regression gates. The platform routes end-user requests to independently built domain agents through the open Agent2Agent (A2A) protocol.
- Led DevOps for the orchestrator, owning Bicep-provisioned Azure Container Apps, Cosmos DB, Key Vault, and multi-environment, approval-gated Azure DevOps CI/CD. Delivered secure infrastructure designed for high availability through Azure Front Door, Application Gateway (WAF), and VNets, alongside end-to-end Azure AD B2C identity propagation, per-tenant isolation, and cross-service OpenTelemetry tracing.
- Member of the AI Center of Excellence; tasked as an advisor to 10+ participating teams in an internal AI hackathon, guiding their solutions on cloud infrastructure and AI/ML design decisions.

Tiger Analytics — Chennai, Tamil Nadu
Senior Analyst - MLE — January 2024 - June 2024

- Delivered a Python-based Retrieval-Augmented Generation (RAG) application with a custom document index for a mobile payments service, improving retrieval accuracy by 20% and lowering response time by 60% through API load balancing and efficient prompt flow design.
- Established a company-wide, Git-based LLMOps workflow in Azure ML SDK for prompt flow experimentation and evaluation, simplifying deployment across Docker images, Kubernetes clusters, and Azure ML endpoints.

Tiger Analytics — Chennai, Tamil Nadu
Analyst - MLE — September 2022 - December 2023

- Enhanced a product portfolio management platform for a global FMCG client by designing automated data ingestion and preprocessing pipelines that integrated high-volume data into SQL databases using Azure Data Factory and SQL-based transformations to deliver application-ready data models.
- Built end-to-end analytics solutions using automated Databricks pipelines and Power BI dashboards with scheduled refresh and KPI computation, improving the reliability and repeatability of business reporting workflows.
- Migrated the codebase and its CI/CD pipelines from Azure DevOps to GitHub Actions, implementing infrastructure-as-code, versioning, and quality checks to strengthen deployment consistency and engineering standards.

Projects

- **Wireless Charging System** *Python, SketchUp, YOLO, OpenCV*
  Worked on a wireless charging system that will charge electric vehicles on the go. Conceptualized and designed the whole charging process which was based on the principles of electromagnetic induction. Implemented Object Detection to create an Autonomous Vehicle Driving System using TensorFlow and OpenCV. Developed an Automatic Number Plate Recognition (ANPR) system to read a vehicle's number plate in real time.
- **Image Segmentation** *Python, NumPy, OpenCV*
  Implemented an 8-neighboring algorithm for image segmentation. Dividing the image into pixel blocks, it compared each block's color features with its 8 surrounding neighbors. Based on similarity scores, colors were assigned to each block, effectively segmenting the image based on color patterns.
- **Semantics-based IR System** *Python, NLP, GloVe Embeddings*
  Built a semantics-driven information retrieval system in Python that preprocesses search queries and documents, assigns relevance scores, and returns the most relevant documents based on semantic similarity rather than keyword matching.

Education

Indian Institute of Information Technology and Management — Trivandrum, Kerala
MSc Computer Science (Machine Intelligence) *GPA: 8.02* — September 2020 - June 2022

Deen Dayal Upadhyaya College — Delhi, India
BSc. Computer Science *GPA: 7.67* — June 2017 - July 2020